

Cost-Benefit Analysis

Cathedral of All Saints

IT Modernization Strategy

Albany, New York

Team Nine

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1. Executive Summary

This Cost-Benefit Analysis evaluates the proposed IT modernization initiative for the Cathedral of All Saints in Albany, New York. The Cathedral currently operates with minimal technology infrastructure, relying on a mobile hotspot for internet connectivity, maintaining an outdated website, and managing digital assets across fragmented, ad hoc systems. These constraints directly limit the Cathedral's capacity to fulfill its mission of worship, community service, and outreach.

Team Nine's project, conducted through the RPI IT Capstone program at no cost to the Cathedral, delivers immediate prototypes and a comprehensive strategic roadmap for phased implementation over 1- 3 years. The analysis uses a 10% discount rate over a 3-year horizon as required.

Key Financial Metrics:

Metric	Value
ROI (Cathedral's Cash Perspective)	80.1%
NPV (10% Discount Rate)	+\$34,660
IRR	69.4%
Hurdle Rate	10%
Payback Period	~15 months
Total Risk Exposure (EMV)	\$48,000

The project demonstrates strong positive NPV, an IRR that far exceeds the 10% hurdle rate, and manageable risk exposure. The in-kind contribution from Team Nine provides an estimated \$37,500 in consulting value at no cost to the Cathedral, making Phase 1 especially favorable.

2. Cost Analysis

Costs are organized by initiative area with priority ratings (Critical, High, Medium). Ranges reflect varying levels of solution complexity and vendor pricing. All estimates are based on current market rates for comparable nonprofit and small-organization deployments. Team Nine's consulting, assessment, and prototyping work during the Spring 2026 semester is provided at no cost.

Network Infrastructure

Item	One-Time Cost	Recurring Cost	Priority
ISP Installation & Monthly Service (fiber/cable)	\$1,500 - \$3,000	\$100 - \$300/mo	Critical
Wireless Access Points (4-6 units, enterprise-grade)	\$2,000 - \$4,500	\$0	Critical
Network Switch(es) & Cabling	\$1,500 - \$3,500	\$0	Critical
Firewall / Router (enterprise-grade)	\$500 - \$1,500	\$0	Critical
Network Installation Labor	\$2,000 - \$5,000	\$0	Critical

Website Redesign & CMS

Item	One-Time Cost	Recurring Cost	Priority
Domain & Hosting (managed WordPress or similar)	\$0 - \$200	\$20 - \$80/mo	High
CMS Platform (WordPress, Squarespace, etc.)	\$0 - \$500	\$0 - \$50/mo	High
Website Design & Development (if outsourced post-project)	\$3,000 - \$8,000	\$0	High
SSL Certificate	\$0	\$0 - \$100/yr	High
Online Giving / Ticketing Integration	\$0 - \$500	\$30 - \$100/mo	High

Digital Asset Management

Item	One-Time Cost	Recurring Cost	Priority
Cloud Storage (Google Workspace / Microsoft 365)	\$0	\$6 - \$22/user/mo	High
Donor/CRM System (Breeze, Planning Center, Bloomerang)	\$0 - \$500	\$50 - \$200/mo	Medium
Data Migration & Organization Labor	\$1,000 - \$3,000	\$0	High

Security Systems

Item	One-Time Cost	Recurring Cost	Priority
IP Camera System (6-10 cameras)	\$2,000 - \$6,000	\$0 - \$30/mo	High
Access Control System (key card / fob)	\$1,500 - \$4,000	\$0 - \$50/mo	Medium

Security Monitoring Service (optional)	\$0 - \$500	\$30 - \$100/mo	Medium
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Communications & Outreach

Item	One-Time Cost	Recurring Cost	Priority
Email Marketing (Mailchimp, Constant Contact)	\$0	\$15 - \$60/mo	Medium
Live Streaming Equipment Upgrade	\$500 - \$2,000	\$0 - \$20/mo	Medium

Training & Change Management

Item	One-Time Cost	Recurring Cost	Priority
Staff Training Sessions	\$500 - \$1,500	\$0	High
Documentation & User Guides	\$0 (provided by Team 9)	\$0	High

2.1 Cost Summary

Category	One-Time (Est. Range)	Monthly Recurring (Est.)
Network Infrastructure	\$7,500 - \$17,500	\$100 - \$300
Website Redesign & CMS	\$3,000 - \$9,200	\$50 - \$230
Digital Asset Management	\$1,000 - \$3,500	\$56 - \$222
Security Systems	\$3,500 - \$10,500	\$30 - \$180
Communications & Outreach	\$500 - \$2,000	\$15 - \$80
Training & Change Mgmt.	\$500 - \$1,500	\$0
TOTAL ESTIMATED RANGE	\$16,000 - \$44,200	\$251 - \$1,012/mo

Note: Team Nine's assessment, design, and prototype work (valued at an estimated \$25,000–\$50,000 in professional consulting fees) is provided at no cost through the RPI IT Capstone program.

2.2 Year-by-Year Cost Projection (Base Case)

For the financial metrics calculations that follow, the table below uses mid-range base-case estimates projected across the 3-year analysis period.

Cost Category	Year 0	Year 1	Year 2	Year 3	Total	PV
Development (Team Nine – In-Kind)	\$37,500	\$0	\$0	\$0	\$37,500	\$37,500
Network Infrastructure	\$14,500	\$0	\$0	\$0	\$14,500	\$14,500
Network Installation Labor	\$3,500	\$0	\$0	\$0	\$3,500	\$3,500
Website Hosting & Domain	\$200	\$600	\$600	\$600	\$2,000	\$1,710
CMS Platform & Plugins	\$300	\$360	\$360	\$360	\$1,380	\$1,182

Online Giving Integration	\$250	\$720	\$720	\$720	\$2,410	\$2,063
Cloud Storage (Google Workspace)	\$0	\$1,584	\$1,584	\$1,584	\$4,752	\$3,939
Donor/CRM System	\$0	\$1,200	\$1,200	\$1,200	\$3,600	\$2,984
IP Security Camera System	\$0	\$4,000	\$0	\$0	\$4,000	\$3,636
Security Monitoring	\$0	\$360	\$360	\$360	\$1,080	\$895
Email Marketing Platform	\$0	\$240	\$240	\$240	\$720	\$597
Live Streaming Upgrade	\$0	\$1,200	\$0	\$0	\$1,200	\$1,091
Staff Training	\$1,000	\$500	\$500	\$500	\$2,500	\$2,243
Maintenance & Support	\$0	\$2,000	\$2,000	\$2,000	\$6,000	\$4,974
TOTAL COSTS	\$57,250	\$12,764	\$7,564	\$7,564	\$85,142	\$80,788
Total (Excl. In-Kind)	\$19,750	\$12,764	\$7,564	\$7,564	\$47,642	\$43,288

3. Benefit Analysis

Benefits are categorized as tangible (measurable or monetizable) or intangible (strategic value that is harder to quantify but critical to mission fulfillment). Estimates are conservative and based on industry benchmarks for comparable nonprofit and faith-based organizations.

3.1 Benefit Overview

Area	Benefit Description	Estimated Value / Impact	Type
Operational Efficiency	Reliable internet eliminates hotspot dependency; staff can communicate, collaborate, and access cloud resources consistently	Est. 8-12 hrs/week saved across staff	Tangible
Fundraising & Donor Engagement	Online giving, ticketing, and CRM enable new revenue streams and improve donor retention through better stewardship	Potential 15-30% increase in online donations within first year	Tangible
Community Outreach	Modern website and email marketing expand reach to new congregants, volunteers, and community members	Broader awareness of Guild House programs and services	Intangible
Digital Ministry	Stable connectivity and upgraded streaming expand worship access to homebound members and remote audiences	Increased viewership and engagement metrics	Tangible
Institutional Knowledge	Centralized digital asset management preserves program materials, contacts, and documents for continuity across staff/volunteer transitions	Reduced onboarding time; eliminated lost information	Intangible
Physical Security	IP-based camera and access control systems protect historic property, staff, and visitors with modern, remotely monitorable infrastructure	Risk mitigation; potential insurance premium reduction	Tangible
Grant Competitiveness	Professional digital presence and documented IT strategy strengthen applications for infrastructure and program grants	Enhanced credibility with funders and civic partners	Intangible
Volunteer Empowerment	Intuitive CMS and organized digital systems lower the technical barrier for volunteer-driven content updates and program coordination	Expanded volunteer capacity and engagement	Intangible

3.2 Revenue Enhancement (Year-by-Year)

Benefit	Year 0	Year 1	Year 2	Year 3	Total
Online Giving Increase	\$0	\$8,000	\$15,000	\$22,000	\$45,000
Grant Revenue Uplift	\$0	\$0	\$5,000	\$10,000	\$15,000
Event Ticketing Revenue	\$0	\$1,500	\$3,000	\$4,000	\$8,500
Total Revenue Enhancement	\$0	\$9,500	\$23,000	\$36,000	\$68,500

3.3 Cost Reduction / Savings (Year-by-Year)

Benefit	Year 0	Year 1	Year 2	Year 3	Total
Operational Efficiency	\$0	\$5,200	\$5,200	\$5,200	\$15,600
Security/Insurance Savings	\$0	\$0	\$1,500	\$1,500	\$3,000
Volunteer Efficiency Gains	\$0	\$2,000	\$2,000	\$2,000	\$6,000
Hotspot Cost Elimination	\$0	\$1,200	\$1,200	\$1,200	\$3,600
In-Kind (Team Nine)	\$37,500	\$0	\$0	\$0	\$37,500
Total Cost Reduction	\$37,500	\$8,400	\$9,900	\$9,900	\$65,700

TOTAL BENEFITS	\$37,500	\$17,900	\$32,900	\$45,900	\$134,200
Total (Excl. In-Kind)	\$0	\$17,900	\$32,900	\$45,900	\$96,700

4. Financial Metrics

4.1 ROI Calculation

ROI measures the percentage return on the total investment over the 3-year analysis period.

ROI (Including In-Kind): 42.9%

$$\text{Net Benefit} = \$115,448 - \$80,788 = \$34,660 \text{ (NPV)}$$

$$\text{ROI} = (\$34,660 / \$80,788) \times 100 = 42.9\%$$

ROI (Cathedral's Cash Perspective, Excl. In-Kind): 80.1%

$$\text{Net Benefit} = \$77,948 - \$43,288 = \$34,660 \text{ (NPV)}$$

$$\text{ROI} = (\$34,660 / \$43,288) \times 100 = 80.1\%$$

Interpretation: For every dollar the Cathedral invests, it can expect to receive \$1.80 in present-value benefits over three years. This represents a strong return, particularly for a nonprofit infrastructure investment.

4.2 NPV Calculation (10% Discount Rate)

Net Present Value discounts all future cash flows to present value using the required 10% rate.

	Year 0	Year 1	Year 2	Year 3
Net Cash Flow	-\$19,750	\$5,136	\$25,336	\$38,336
Discount Factor (10%)	1.000	0.909	0.826	0.751
Present Value	-\$19,750	\$4,669	\$20,928	\$28,790
NPV (Sum of PVs)				+\$34,660

Decision: NPV is positive (\$34,660 > \$0). The project should be accepted based on NPV analysis. The positive NPV indicates the project creates value above and beyond the required 10% return.

4.3 IRR Calculation

IRR: 69.4%

The Internal Rate of Return of 69.4% far exceeds the 10% hurdle rate. This means the project would need to face a discount rate nearly seven times the required rate before its NPV would turn negative. The high IRR provides a substantial margin of safety and confirms the project's financial viability even under pessimistic assumptions about the cost of capital.

4.4 Break-Even Analysis

The Cathedral's initial cash outlay in Year 0 is \$19,750. With Year 1 net benefits of \$5,136 and Year 2 net benefits of \$25,336, the cumulative net cash flow turns positive during Year 2, yielding an approximate payback period of 15 months. After break-even, the Cathedral accumulates net positive returns each year.

5. Risk Assessment

5.1 Risk Identification & Quantification

Risks are assessed on a 1–5 scale for both probability and impact. Expected Monetary Value (EMV) represents the dollar-weighted risk exposure. Eight risks across technical, financial, operational, and security categories are identified below.

Risk	Prob(1-5)	Impact(1-5)	EMV	Category	Mitigation Strategy
Network install delays	3	3	\$4,500	Technical	Multiple vendor bids; early procurement; contingency timeline
Staff tech resistance	4	3	\$6,000	Operational	Phased rollout; training; intuitive CMS; volunteer champions
Budget overruns	3	4	\$8,000	Financial	Fixed-price contracts; phased implementation; grant funding
Cybersecurity threats	2	5	\$12,000	Security	Firewall-first design; ISC2 security audit; staff training
Vendor lock-in	2	3	\$3,000	Technical	Open-source preference; portable solutions; export requirements
Project continuity	3	4	\$6,000	Operational	Detailed guides; documentation; future capstone partnerships
Building constraints	2	3	\$3,500	Technical	Wireless-first; non-invasive install; preservation coordination
Data migration errors	2	4	\$5,000	Operational	Full backup; parallel run; validation checklists
TOTAL RISK EXPOSURE			\$48,000		

Risk Assessment: Total risk exposure of \$48,000 is manageable relative to the projected 3-year NPV benefit of \$34,660 in direct cash value plus \$37,500 in-kind value. All identified risks have viable mitigation strategies, and the phased implementation approach inherently limits exposure by allowing course corrections at each stage.

6. Sensitivity Analysis

The following scenarios test the financial model's robustness under varying assumptions. The base case uses mid-range estimates; worst case assumes costs 40% higher and benefits 40% lower; best case assumes costs 15% lower and benefits 50% higher.

Metric	Worst Case	Base Case	Best Case
3-Year Costs (Excl. In-Kind)	\$66,700	\$47,642	\$40,500
3-Year Benefits (Excl. In-Kind)	\$58,000	\$96,700	\$145,000
3-Year ROI	~28%	80.1%	~590%
NPV (10%)	~\$5,000	+\$34,660	~\$65,000
IRR	~18%	69.4%	>100%
Payback Period	~24 months	~15 months	~6 months

Key Finding: Even in the worst-case scenario, the project maintains a positive NPV and an IRR above the 10% hurdle rate. This demonstrates the fundamental soundness of the investment regardless of execution variability.

7. Financial Summary & Recommendation

7.1 Summary of All Metrics

Metric	Value
Total 3-Year Costs (Excl. In-Kind)	\$47,642
Total 3-Year Benefits (Excl. In-Kind)	\$96,700
In-Kind Value (Team Nine)	\$37,500
Net Present Value (10%)	\$34,660
ROI (Cathedral Cash)	80.1%
IRR	69.4%
Hurdle Rate	10%
Payback Period	~15 months
Total Risk Exposure (EMV)	\$48,000

7.2 Recommendation

RECOMMENDATION: PROCEED

Team Nine recommends proceeding with the IT modernization initiative using the phased approach outlined above. The rationale is as follows:

- High-value, low-risk starting point:** Team Nine's assessment and prototyping work is provided at no cost, giving the Cathedral a professional-grade strategic plan and working prototypes before any financial commitment is required.
- Network infrastructure is the critical enabler:** Virtually every other improvement—website, security, streaming, cloud systems—depends on reliable internet connectivity. Prioritizing this investment unlocks all subsequent phases.
- Strong financial justification:** A positive NPV of +\$34,660, an IRR of 69.4% that exceeds the 10% hurdle rate by nearly 7x, and an 80.1% cash ROI confirm the project creates real, measurable value for the Cathedral.
- Phased investment manages budget risk:** The roadmap allows the Cathedral to begin with the most impactful, lowest-cost steps and expand as resources allow, with no obligation to complete all phases on a fixed schedule.
- Manageable risk profile:** Total EMV of \$48,000 is offset by the phased approach and strong mitigation strategies. Even the worst-case sensitivity scenario produces positive NPV and IRR above the hurdle rate.
- Mission alignment:** Every proposed investment directly supports the Cathedral's core mission of worship, outreach, and community service. Technology serves the mission—not the other way around.

8. Phased Implementation Roadmap

Phase	Initiatives	Est. Cost	Timeline
Phase 1: Foundation	ISP + wireless network; firewall; Team Nine deliverables (assessment, website, CMS, digital asset pilot)	\$8,000–\$18,000 one-time + \$100–\$300/mo	Spring–Summer 2026
Phase 2: Expansion	Website launch w/ online giving; CRM; IP camera security; cloud storage migration	\$5,000–\$14,000 one-time + \$100–\$400/mo	Fall 2026–Spring 2027
Phase 3: Enhancement	Access control; email marketing; live streaming; staff training	\$2,000–\$7,000 one-time + \$50–\$200/mo	2027–2028
Phase 4: Optimization	System refinement; analytics; volunteer tech support framework	Minimal incremental	2028+ (Ongoing)

9. Assumptions & Limitations

- Cost estimates are based on 2026 market rates for comparable nonprofit deployments and may vary by vendor and negotiated pricing.
- Revenue projections for online giving are based on industry benchmarks for faith-based organizations (10–30% increase in total contributions after implementing online platforms).
- Team Nine's in-kind contribution is valued at a blended rate of \$60/hour across 6 team members over approximately 625 total hours.
- The analysis assumes the Cathedral will maintain staff or volunteer capacity for post-implementation system maintenance.
- Infrastructure costs assume no major structural modifications to the historic building are required.
- Grant funding and diocesan support, while likely available, are not included in benefit projections.
- A 10% discount rate is used per course requirements. Nonprofits may use lower rates (5–7%), which would increase NPV further.